

FRESHMAN SEMINAR · ANSWER KEY - PROFESSOR COPY

Homework 1: Notes Practice

Grading guide. Total **21 points**. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Reading your timetable

1. What time is your Biology 101 lecture on Monday? And when is the Biology recitation? (2 pts)

Answer: Lecture Monday 1:00; recitation Tuesday 9:45.

2. Which class comes right after the Chemistry recitation on Tuesday? (1 pt)

Answer: The Biology recitation (Tuesday 9:45).

Part B — Three big ideas (the main event)

From NICU Tutor: The pulse oximeter is a soft band with two tiny lights – one red, one infrared – that shine right through a baby's foot. Hemoglobin that is carrying oxygen absorbs those two colors differently from hemoglobin that is empty. By comparing how much of each light gets through, the machine works out what percentage of the hemoglobin is carrying oxygen. Because every heartbeat pushes a fresh pulse of blood into the foot, the machine can also count the heart rate. For preemies getting extra oxygen, the team aims for about 90 to 95 percent, because too much oxygen can damage the eyes and lungs.

3. Write the **THREE** biggest ideas from that passage, in your own words. One line each. (2 points each: the idea is right and it isn't just copied.) (6 pts)

Answer: Any three of: (1) It measures blood oxygen by shining two colors of light through the foot. (2) Loaded and empty hemoglobin absorb light differently – that's the trick. (3) It also reads heart rate from the pulse of each heartbeat. (4) Preemies on oxygen are kept around 90-95% because too much oxygen harms eyes/lungs. Give 1 point if the idea is right but copied word-for-word.

4. Two students took notes on that passage. Student 1 wrote: "The pulse oximeter is a soft band with two tiny lights, one red, one infrared, that shine right through a baby's foot." Student 2 wrote: "Pulse ox = shines 2 light colors through foot to measure oxygen." Which student followed the note-taking rule, and why? (2 pts)

Answer: Student 2 – the note is in their own words and captures the idea (what it does and how), not a copied sentence. Student 1 copied a sentence, which doesn't prove understanding and takes longer.

Part C — New words

5. Pick TWO new words from any of today's classes (Seminar, Chemistry, or Biology). Write each word, then what it means in your own words. (4 pts)

Answer: Any two real terms from the day with a reasonable definition (e.g., molecule — atoms joined together; recitation — where we practice/do homework; mitochondria — cell's power plants). 1 point for the word, 1 for the meaning.

Part D — Quiz strategy

6. You've read a quiz question twice and crossed out two choices you know are wrong. What's the best next step? (1 pt)

Answer: A. Compare the two remaining choices carefully against what the question actually asks. - Two left is a fair fight — reread the question and check each against it. Length and position tell you nothing here.

Part E — Flashback to NICU Tutor

7. What does the skin probe taped to a baby's belly in an isolette do? (1 pt)

Answer: Measures the baby's skin temperature so the isolette can adjust its warmth automatically (servo control).

8. Name TWO of the four ways a baby loses heat. (2 pts)

Answer: Any two of: evaporation, convection, radiation, conduction.

Part F — Think like a student

9. Elizabeth Blackwell was let into medical school as a joke and graduated first in her class. Think about your own week: name ONE specific thing you can do if a class feels hard. (2 pts)

Answer: Any thoughtful, specific answer: ask the professor (office hours), reread the 'Write this down' boxes, use the read-aloud, retake the quiz after reviewing, look back at the NICU Tutor lesson, take a break and come back. 2 points for specific; 1 for vague ('try harder').